

# Amorphous crystals: Supplementary Material

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*a. Fundamental and elastic domains.* We refer to the restriction of the energy density to the (minimal periodicity) fundamental domain as  $\varphi_D(\tilde{\mathbf{C}})$ , where  $\tilde{\mathbf{C}} = \mathbf{m}^T \mathbf{C} \mathbf{m}$  is the projection of a general metric tensor  $\mathbf{C}$  into this domain and  $\mathbf{m}$  is the corresponding mapping of the metric tensor  $\mathbf{C}$  into the fundamental domain ensuring that  $\varphi(\mathbf{C}) = \varphi_D(\tilde{\mathbf{C}})$ . In our 2D case the form of the fundamental domain is known [1–4]

$$D = \left\{ 0 < C_{11} \leq C_{22}, \quad 0 \leq C_{12} \leq \frac{C_{11}}{2} \right\}. \quad (1)$$

where  $\det \mathbf{C} = 1$ . Given a generic metric  $\tilde{\mathbf{C}}$ , the task of finding a unimodular matrix  $\mathbf{m}$  ensuring that  $\tilde{\mathbf{C}} \in D$  (and therefore  $\varphi(\mathbf{C}) = \varphi_D(\tilde{\mathbf{C}})$ ) can be formulated in the form of a recursive algorithm which is also well known (Lagrange reduction) [2, 3]. Specifically, if we define the matrices

$$\mathbf{m}_1 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \mathbf{m}_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \mathbf{m}_3 = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}. \quad (2)$$

and start with an assumption that  $\mathbf{m} = \mathbb{I}$ , we can proceed through the following steps: (i) if  $C_{12} < 0$ , change sign of  $C_{12}$  using the mapping  $\mathbf{m} \rightarrow \mathbf{m}\mathbf{m}_1$ ; (ii) if  $C_{22} < C_{11}$ , swap these two components using the mapping  $\mathbf{m} \rightarrow \mathbf{m}\mathbf{m}_2$ ; (iii) if  $2C_{12} > C_{11}$ , set  $C_{12} = C_{12} - C_{11}$  using the mapping  $\mathbf{m} \rightarrow \mathbf{m}\mathbf{m}_3$ . Note that the action of the matrix  $\mathbf{m}_1$  is a reflection which returns an acute angle between two lattice vectors of the square lattice  $\mathbf{e}_i$  where  $i = 1, 2$ ; the action of the matrix  $\mathbf{m}_2$  is also a reflection as it swaps two lattice vectors  $\mathbf{e}_i$ . Both of these operations belong to the point group and propagate the metric only inside the corresponding ‘elastic domain’ (Ericksen-Pitteri neighborhood) composed of the four copies of the minimal domain  $D$  [4, 5]. Instead, the mapping defined by matrix  $\mathbf{m}_3$  brings the metric outside the ‘elastic domain’ and therefore represents a quantized analog of the macroscopic plastic strain.

*b. Elastic energy density.* While the single-period Landau potential  $\varphi_D(\tilde{\mathbf{C}})$  can be constructed using the classical Cauchy-Born approach, see for instance [6, 7], in this paper we used for simplicity the phenomenological expression proposed in [2]. Specifically, the elastic energy density is represented as a sum of two terms:

$$\varphi(\tilde{\mathbf{C}}) = \varphi_0 \left( \frac{\tilde{\mathbf{C}}}{(\det \tilde{\mathbf{C}})^{1/2}} \right) + \varphi_v(\det \tilde{\mathbf{C}}), \quad (3)$$

where  $\varphi_0$  accounts for contributions due to shear while  $\varphi_v$  penalizes volumetric deformations. The  $GL(2, \mathbb{Z})$  invariant shear term is chosen in the form sixth order polynomial which is the minimal requirement ensuring stress continuity over the

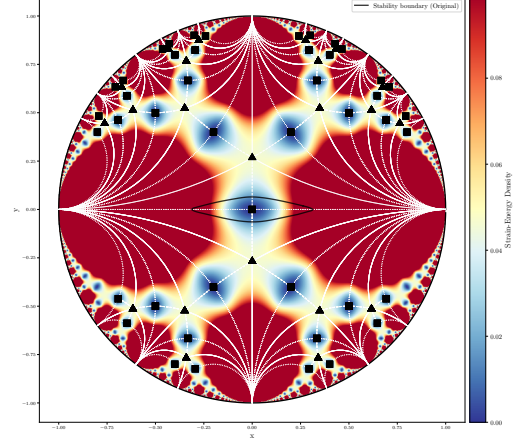


FIG. 1. Strain-energy density landscape in the space of shear strains visualized on a Poincaré disk using stereographic projection. Thin white lines show the original tessellation into equivalent periodicity domains. Colors represent level of energy. The stability boundaries obtained from the equation (10) are shown by a thick black line around the reference energy well.)

whole configurational space [1, 2]. It depends on a single parameter  $\beta$  allowing one to associate ground state either with square or triangular (hexagonal) lattice

$$\varphi_0 \left( \frac{\tilde{\mathbf{C}}}{(\det \tilde{\mathbf{C}})^{1/2}} \right) = \beta \psi_1 \left( \frac{\tilde{\mathbf{C}}}{(\det \tilde{\mathbf{C}})^{1/2}} \right) + \psi_2 \left( \frac{\tilde{\mathbf{C}}}{(\det \tilde{\mathbf{C}})^{1/2}} \right). \quad (4)$$

The requirement of stress continuity across the whole periodic energy landscape specifies the functions  $\psi_1$  and  $\psi_2$

$$\psi_1 = I_1^4 I_2 - \frac{41}{99} I_2^3 + \frac{7}{66} I_1 I_2 I_3 + \frac{1}{1056} I_3^2, \quad (5)$$

$$\psi_2 = \frac{4}{11} I_2^3 + I_1^3 I_3 - \frac{8}{11} I_1 I_2 I_3 + \frac{17}{528} I_3^2. \quad (6)$$

where we used the known hexagonal invariants of the metric tensor

$$I_1 = \frac{1}{3} (\tilde{C}_{11} + \tilde{C}_{22} - \tilde{C}_{12}), \quad (7)$$

$$I_2 = \frac{1}{4} (\tilde{C}_{11} - \tilde{C}_{22})^2 + \frac{1}{12} (\tilde{C}_{11} + \tilde{C}_{22} - 4\tilde{C}_{12})^2, \quad (8)$$

$$I_3 = (\tilde{C}_{11} - \tilde{C}_{22})^2 (\tilde{C}_{11} + \tilde{C}_{22} - 4\tilde{C}_{12}) - \frac{1}{9} (\tilde{C}_{11} + \tilde{C}_{22} - 4\tilde{C}_{12})^3. \quad (9)$$

We select  $\beta = -1/4$  to ensure that global energy minimizers correspond to square lattices. The volumetric part of the energy density, which primarily influences the fine structure of

the dislocation cores and also controls the formation of voids, is assumed to be of a generic form preventing infinite compression  $\varphi_v(s) = \mu(s - \log(s))$ . To keep in our numerical experiments the strain field close the surface  $\det C = 1$ , we adopted a sufficiently high value of the bulk modulus,  $\mu = 5$ .

The isochoric part of the energy landscape is illustrated in Fig. 2 of the main paper. Here we also show the same Landau-type potential in Fig. 1, but now in the level set representation. Dark blue regions correspond to the equivalent square energy wells. Red domains show effectively inaccessible regions where the energy is very high (truncated). Relatively low energy valleys colored in yellow correspond roughly to two simple shear plastic 'mechanisms' available in square lattices.

*c. Internal length scale.* The lack of convexity of the potential is a property which the MTM shares with other similar Landau-type continuum theories. In the MTM approach the necessary regularization is introduced by bringing into the model an internal scale  $h$  through explicit spatial discretization. If  $L$  is the linear size of the macroscopic domain and  $N^2$  is the number of the nodes in the mesoscopic grid, then the parameter  $h = L/N$  introduces elastic finite elements imitating mesoscopic aggregates of atomic particles and defines in this way the cutoff beyond which the deformation is considered homogeneous (affine). The corresponding small parameter is  $\delta = h/L = 1/N$ . Another small parameter in the problem is  $\epsilon = a/L$  where  $a$  is the interatomic distance and we are interested in the double limit:  $\epsilon \rightarrow 0$  and  $\delta \rightarrow 0$  while  $\delta \gg \epsilon$ . In the MTM approach we first implicitly perform the limit  $\epsilon \rightarrow 0$  and recover in this way the continuum constitutive response using the Cauchy-Born rule. Then instead of performing the classical  $\delta \rightarrow 0$  and obtaining the conventional scale-free continuum theory, we preserve in the theory small but finite value of the parameter  $\delta$ . In this way we effectively account for finite size effects. In particular, the size of the dislocation cores is overestimated while the number of dislocations is underestimated. However, the resulting approach preserves the basic nature of both long range and short range interaction of dislocations at least when they are away from the boundaries. For instance the blown up dislocations will still nucleate and self-lock adequately. In our simulations we adopted a pragmatic approach to the choice of the scale  $h$  by requiring that the Cauchy-Born energy density, computed for a lattice fragment with scale  $h$ , maintains periodicity within the interesting range of strains.

*d. Numerical method.* Numerical implementation of the MTM approach reduces to solving an elastic finite element problem. The goal is to follow the displacements of the network of discrete nodes labeled by integer valued coordinates  $I = 1, \dots, N^2$ .

Specifically, we assume that each of the elements is a deformable triangle and employ the standard linear 3-node elements [8]. The displacement field inside each of the elements can be written in the form  $\mathbf{u}(\mathbf{z}) = \sum_a \mathcal{N}^a(\mathbf{z}; h) \mathbf{u}^a$ , where  $\mathcal{N}^a(\mathbf{z}; h)$  are compactly supported linear shape functions,  $\mathbf{u}^a$  are the amplitudes of the displacements of the nodes and summation is assumed over repeated indexes; the interpolation functions for each element are defined in terms of local

dimensionless coordinate system. The mesoscopic deformation gradient is then  $\mathbf{F}(\mathbf{z}; h) = \mathbb{I} + \mathbf{u}^a \otimes \nabla \mathcal{N}^a(\mathbf{z}; h)$ .

In terms of macroscopic reference coordinates, the elastic energy inside each element can be computed using the simplest quadrature scheme  $w(\mathbf{x}; h) = \frac{1}{2} \varphi(\mathbf{F}(\mathbf{x}; h)) J(\mathbf{x}, h)$ , where  $J$  is the Jacobian of the transformation from local ( $\mathbf{z}$ ) to global ( $\mathbf{x}$ ) coordinate system. Note that in view of our assumptions the energy of a deformed triangle element depends on three parameters: the deformed lengths of the bonds and the deformed value of the angle between them.

Finding solution of an elastic problem implies local minimization of the energy  $W = \int_{\Omega} w(\mathbf{x}; h) d\mathbf{x}$ , which is prescribed on a triangulated domain  $\Omega$ . The conditions of mechanical equilibrium, formulated in terms of the first Piola-Kirchhoff stress tensor  $\mathbf{P} = 2\mathbf{F} \mathbf{m} \frac{\partial w}{\partial \mathbf{C}} \mathbf{m}^T$ , take the form  $\nabla \cdot \mathbf{P} = 0$ . In the finite element representation these equations reduce to  $\frac{\partial W}{\partial \mathbf{u}^a} = \int_{\Omega} \mathbf{P}(\mathbf{x}; h) \nabla \mathcal{N}^a(\mathbf{x}; h) d\mathbf{x} = 0$ .

The ensuing mathematical problem is solved numerically using the L-BFGS algorithm [9] which builds a positive definite linear approximation allowing one to make a quasi-Newton step lowering the total energy. The iterations continue till the increment in the advance of the total energy becomes sufficiently small.

More specifically, an approximate solution is used as an initial guess  $\mathbf{w}^a$  to solve, using LU factorization [10, 11], the linear equations for the correction  $d\mathbf{w}^a = \mathbf{u}^a - \mathbf{w}^a$  which are of the form  $K_{ij}^{ab} dw_j^b + R_i^a = 0$ , where  $K_{ij}^{ab} = A_{ipjq}(\mathbf{F}) \frac{\partial \mathcal{N}^a}{\partial x_p} \frac{\partial \mathcal{N}^b}{\partial x_q}$  and  $R_i^a = P_{ip}(\mathbf{F}) \frac{\partial \mathcal{N}^a}{\partial x_p}$ . Here we also defined the tensor of tangential elastic moduli  $A_{iajb} = \frac{\partial^2 \varphi_D(\tilde{\mathbf{C}})}{\partial F_{ia} \partial F_{jb}}$ . Note that using the Eulerian acoustic tensor  $Q_{ij} = F_{la} F_{mb} A_{iajb} n_l n_m$ , where  $\mathbf{n}$  is a unit vector we can obtain the Legendre-Hadamard threshold of elastic instability

$$\det(\mathbf{Q}) = 0, \quad (10)$$

which was used in the main paper and which is illustrated for our chosen elastic energy density in Fig. 1.

*e. Implementation of the loading conditions.* To simulate an effectively infinite crystal under controlled macroscopic affine deformation, we use the method of domain replication. Starting from the computational domain  $\Omega_0$  with dimensions  $L_x \times L_y$  we generate a periodic lattice of domains using translation vectors  $\mathbf{t}_n = \begin{pmatrix} n_x L_x \\ n_y L_y \end{pmatrix}$ , where  $\mathbf{n} = (n_x, n_y) \in \{-1, 0, 1\}^2$ . Let  $\bar{\mathbf{F}}(\alpha)$  denote the macroscopic deformation gradient tensor at the parameter value  $\alpha$ . Under such affine deformation the position of replicated domains relative to the central domain  $\Omega_0$  is given by:  $\mathbf{R}_n = \bar{\mathbf{F}}(\alpha) \cdot \mathbf{t}_n$ , where  $\mathbf{R}_n$  represents the displacement vector of the origin of the domain associated with  $\mathbf{t}_n$ . Then, for a node located at position  $\mathbf{x}_i$  in the central domain  $\Omega_0$ , its periodic image, associated with the translation vector  $\mathbf{t}_n$ , is at:  $\mathbf{x}_i^{(n)} = \mathbf{x}_i + \mathbf{R}_n = \mathbf{x}_i + \bar{\mathbf{F}}(\alpha) \cdot \mathbf{t}_n$ . Such correspondence ensures that the macroscopic deformation gradient  $\bar{\mathbf{F}}(\alpha)$  is consistently applied across all periodic images of  $\Omega_0$ . After establishing the positions of all nodes across the nine domains (original domain  $\Omega_0$  and its eight periodic images), we per-

form a Delaunay triangulation on the complete set of particles:  $\mathcal{P} = \{\mathbf{x}_i : i \in \mathcal{I}_0\} \cup \{\mathbf{x}_i^{(n)} : i \in \mathcal{I}_0, n \neq \mathbf{0}\}$ , where  $\mathcal{I}_0$  denotes the set of particle indices in the original domain and  $\mathbf{0} = (0, 0)$  corresponds to the null translation vector. using the obtained global triangulation  $\mathcal{T}$ , we select the triangles that have at least one vertex belonging to the original domain  $\Omega_0$ . The initial selection of all triangles with at least one node connected to the original domain may introduce duplicated triangles due to the periodic replication. Therefore, we apply an additional filtering step to remove these duplicates and retain only unique triangles. Such selection procedure ensures that:

- (i) Each triangle in the computational mesh is counted exactly once, eliminating redundancy from the periodic extension;
- (ii) All triangles connected to the original domain are included, maintaining proper connectivity across periodic boundaries;
- (iii) The triangulation naturally incorporates the macroscopic deformation through the deformed positions of the periodic images. The selected set of triangles  $\mathcal{T}_{\text{unique}}$  forms the computational mesh for the MTM calculations while implementing the periodic boundary conditions and ensuring the consistency with the applied macroscopic deformation gradient  $\bar{\mathbf{F}}(\alpha)$ .

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